

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built the scalable document search engine (*Python FastAPI*) of a competitive high school debate research SaaS platform on *AWS* from design through launch, with an *Agile* team of **10 engineers** and **10K+ site visitors**
- Scaled performant search to **600K+ debate documents** on *Elasticsearch*, returning results in **under 1 second**
- Designed a text-retrieval algorithm that validates source URLs and filters out statistical outliers with **IQR**
- Developed a de-duplication algorithm that flagged **30%** of evidence documents as identical, reducing index bloat

Product Development Intern | Jetson

January 2025 – February 2025

- Worked closely with product on **7 iterations of UI/UX testing** across 3 features of the Jetson App (100K+ users)
- Turned customer needs into a **5-page report** recommending ways to optimize user engagement for Jetson's CEO
- Redesigned onboarding on *Figma*, cutting it from **12 to 5 pages**, validated via **A/B testing** on 30 test users

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed 3 years of TikTok data (**200+ posts**) with *Python (pandas)*, *Hootsuite*, *Power BI* and *Microsoft Excel*
- Identified that “science fun facts” posts comprised 69.3% of views, despite accounting for 7% of all content
- Pitched a “fun facts” strategy to the Executive Director, who adopted it; engagement rose **24% (8K+ views)**

PROJECTS

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Built a dual-API ingestion pipeline resolving papers via *OpenAlex* & backfilling metadata from *Semantic Scholar*
- Stored paper data on *Supabase (PostgreSQL)* and wrote *SQL* queries to filter papers by field, year and citation count
- Deployed a **RAG agent** (*Claude Sonnet 4.5*) that retrieves session-scoped vector documents (*Supernemory*)
- Created a citation graph (*React Flow*) that maps a paper's outbound citations across **26 research fields**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Engineered a **suite of 3 edge AI tools** that collects data regarding a school's energy, water and food consumption
- Developed a *Python FastAPI* backend that ingests live water, waste & energy data from all 3 tools, runs **4 Random Forest models** to detect anomalies & deliver actionable steps via **Gemma Voice LLM** on a *Next.js* dashboard
- Led a team of 5 across project ideation, development and presentation. **Finalist at USAII Global Hackathon**

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Designed a **5-step Python scheduling algorithm** that scans calendar availability, returns a ranked list of time slots and sends email confirmation for each booking, validated across **14 unit and integration test cases**
- Built a **scheduling agent** (*React/Firebase*) that auto-books meetings for large groups (30+ ppl) on Google Calendar
- Created a lobby system that enables the host to authenticate invited participants and queue follow-up meetings
- Licensed Chronos to **8 startups** & signed **2 LOIs worth \$3.5K**. **Won 2025 Congressional App Challenge**

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Led a team of 4 on a pricing algorithm for **Kamion (YC S22)**. **2nd Place at UIUC Founder's Hackathon**
- Designed a *JSON schema* for a **Claude Sonnet 4.6 agent** to extract condition, brand, model and year from truck photos, revising it after each wrong prediction until **10 consecutive images** were correct on each of **3 features**
- Built a *quantile regression* pricing model on the extracted features, reaching **87% accuracy** on **30 samples**

SKILLS

Programming Languages: Python, SQL, Java, JavaScript, C++, HTML/CSS

Frameworks/Platforms: Flask, FastAPI, PostgreSQL, AWS, Elasticsearch, MySQL, Supabase, Node.js, React, Next.js

DevTools: Git, GitHub, Docker, CI/CD, Claude Code, Codex, VS Code, Figma, Bun, Catch2, Clang, GDB Debugger

AI/ML: scikit-learn, PyTorch, Keras, TensorFlow, Kaggle, OpenCV RAG, AI Agents, LLMs, AI Automations

Concepts/Practices: Backend Development, Object Oriented Programming, System Design, FinTech, Full Stack